B.TECH 1ST YEAR | ENGINEERING PHYSICS

Engineering Physics

Comprehensive Study Module

Unit III

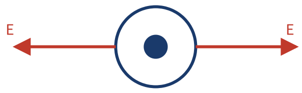
Dielectric & Magnetic Materials

Dielectric Polarization • Electric Susceptibility • Dielectric Constant

Lorentz Internal Field • Clausius–Mossotti Equation • Dielectric Loss

Magnetic Dipole Moment • Magnetization • Classification of Magnetic Materials

Domain Theory • Hysteresis Curve • Soft & Hard Magnetic Materials



Prepared as per Anna University / JNTU / VTU Syllabus | Exam-Oriented Content with Diagrams, Derivations & Q&A

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Symbols and Notations Used

Symbol	Quantity	SI Unit
E	Electric Field Intensity	$V\ m^{-1}$
D	Electric Displacement Vector	$C\ m^{-2}$
P	Polarization Vector	$C\ m^{-2}$
ε_0	Permittivity of Free Space = $8.854 \times 10^{-12}\ F\ m^{-1}$	$F\ m^{-1}$
ε_r	Relative Permittivity (Dielectric Constant)	Dimensionless
χ_e	Electric Susceptibility	Dimensionless
α	Polarizability	$F\ m^2$
H	Magnetic Field Intensity	$A\ m^{-1}$

B	Magnetic Flux Density	T (Tesla)
M	Magnetization	A m^{-1}
μ_0	Permeability of Free Space = $4\pi \times 10^{-7}$ H m^{-1}	H m^{-1}
μ_r	Relative Permeability	Dimensionless
χ_m	Magnetic Susceptibility	Dimensionless
N	Number Density of dipoles	m^{-3}
p	Dipole Moment	C m
m	Magnetic Dipole Moment	A m^2

SECTION 1

Introduction to Dielectric Materials

1.1 What is a Dielectric Material?

Definition: A dielectric material is an electrical insulator that can be polarized by an applied electric field. When placed in an external electric field, the bound charges within the material are displaced slightly from their equilibrium positions, creating electric dipoles throughout the bulk of the material.

Unlike conductors, dielectrics do not allow free flow of electric charges. However, they respond to an external electric field through the process of *polarization* — the microscopic displacement of positive and negative charge centres within atoms or molecules. This internal redistribution of charge alters the macroscopic electric field within the material.

Dielectric materials are fundamental to the operation of capacitors, insulators in transmission lines, substrates in printed circuit boards, and optical waveguides. The study of dielectrics bridges electrostatics and materials science.

1.2 Dielectrics vs Conductors

Property	Conductor	Dielectric (Insulator)
Charge carriers	Free electrons (mobile)	Bound charges (immobile)
Response to E field	Surface charge redistribution	Polarization throughout volume
Internal E field	Zero in equilibrium	Reduced, but non-zero
Resistivity	10^{-8} to 10^{-6} Ω m	10^{8} to 10^{16} Ω m
Examples	Copper, Aluminium, Silver	Glass, Mica, $BaTiO_3$, PTFE
Capacitor use	Plates	Filling between plates

1.3 Electric Dipole and Dipole Moment

An **electric dipole** consists of two equal and opposite charges +q and -q separated by a distance d. The electric dipole moment is a vector quantity pointing from the negative charge to the positive charge.

ELECTRIC DIPOLE MOMENT

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p      = q × d    (vector, from -q to +q)
|p|    = q d    [SI unit: C m (coulomb-metre)]
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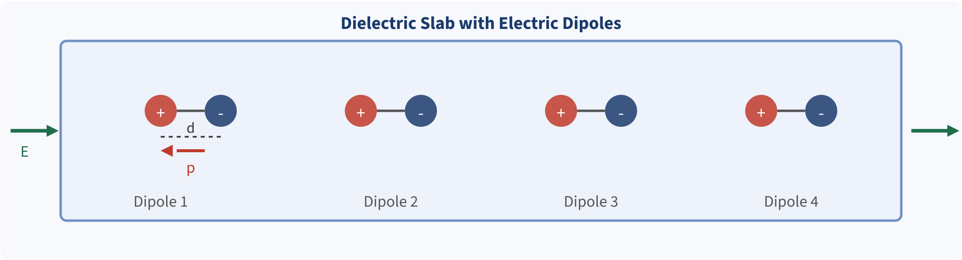

Fig. 1.1 – Electric dipoles aligned in a dielectric slab under applied electric field E. Each dipole has moment p = q d directed from −q to +q.

1.4 Applications of Dielectric Materials

- Capacitor dielectrics: mica, ceramic ($BaTiO_3$), electrolytic oxide layers
- Transformer and cable insulation: polyethylene, oil-impregnated paper
- Piezoelectric sensors and actuators: quartz, PZT (lead zirconate titanate)
- Optical fibres and photonic devices: silica glass, lithium niobate
- Microwave substrates: PTFE (Teflon), alumina ceramics
- Electret microphones: permanently polarized PVDF films

SECTION 2

Dielectric Polarization Concepts

2.1 Polarization of a Dielectric

Polarization (P): Polarization is defined as the electric dipole moment per unit volume of a dielectric material. It represents the degree of alignment of microscopic dipoles in response to an applied electric field.

POLARIZATION VECTOR

P	$= N \times p$ (N = number density of dipoles, p = dipole moment)
P	$= \varepsilon_0 \chi_e E$ (χ_e = electric susceptibility)
Unit	$C\ m^{-2}$ (same as surface charge density)

When an external electric field is applied to a dielectric, charges are displaced from their equilibrium positions. The collective effect of all microscopic displacements is measured by the polarization vector P, which has the same direction as the applied field E in isotropic materials.

2.2 Dielectric Polarizability (α)

Polarizability (α): The polarizability of a molecule or atom is the proportionality constant between the induced dipole moment and the local electric field acting on it.

POLARIZABILITY

p	$= \alpha E_{local}$
α	$= p / E_{local}$ [SI unit: $F\ m^2$ or $C^2\ s^2\ kg^{-1}$]

Polarizability quantifies how easily the electron cloud of an atom or bond is distorted by an electric field. Larger atoms with more diffuse electron clouds have higher polarizability. The total polarizability has contributions from electronic, ionic, and orientational mechanisms.

2.3 Electric Susceptibility (χ_e)

Electric Susceptibility (χ_e): Electric susceptibility is a dimensionless quantity that measures how easily a dielectric material is polarized in response to an applied electric field.

ELECTRIC SUSCEPTIBILITY RELATIONS

$P = \varepsilon_0 \chi_e E$

$\chi_e = \varepsilon_r - 1$ (ε_r = relative permittivity)

$\chi_e > 0$ always for dielectrics (polarization is in same direction as E)

2.4 Dielectric Constant (ε_r)

Dielectric Constant (ε_r): Also called relative permittivity, it is the ratio of the permittivity of the dielectric medium to the permittivity of free space. It represents the factor by which the electric field is reduced inside the dielectric compared to vacuum.

DIELECTRIC CONSTANT

$\varepsilon_r = \varepsilon / \varepsilon_0$

$\varepsilon_r = 1 + \chi_e$

$\varepsilon_r = C / C_0$ (ratio of capacitance with and without dielectric)

$\varepsilon_r \geq 1$ always; vacuum: $\varepsilon_r = 1$; water: $\varepsilon_r \approx 80$; $BaTiO_3$: $\varepsilon_r \approx 1500$

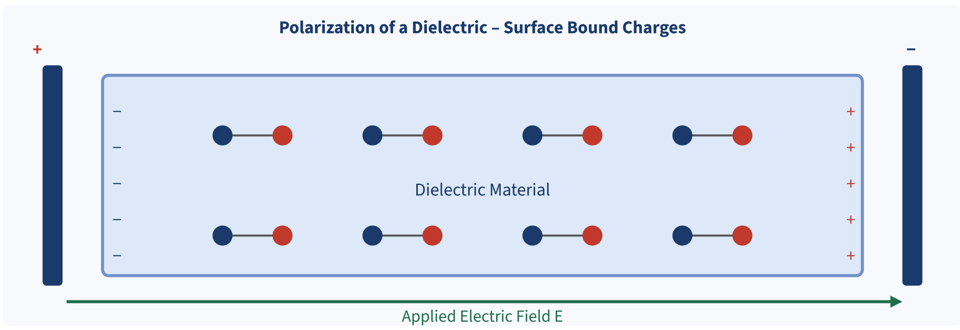

Fig. 2.1 – Dielectric polarization: aligned dipoles create bound surface charges opposing the applied field E.

SECTION 3

Electric Vectors and Relations

3.1 Electric Displacement Vector (D)

Electric Displacement (D): The electric displacement vector D is defined in such a way that it depends only on free charges and is independent of the nature of the dielectric medium. It is also called the electric flux density.

ELECTRIC DISPLACEMENT VECTOR

D	$= \varepsilon_0 E + P$	(General relation in dielectric)
D	$= \varepsilon_0 \varepsilon_r E = \varepsilon E$	(For linear, isotropic dielectric)
Unit	$C\ m^{-2}$	

3.2 Relation between Electric Vectors

In a linear isotropic dielectric, the three electric vectors E, P, and D are related as follows:

COMPLETE SET OF RELATIONS

P	$= \varepsilon_0 \chi_e E$
D	$= \varepsilon_0 E + P$
D	$= \varepsilon_0 E + \varepsilon_0 \chi_e E = \varepsilon_0 (1 + \chi_e) E$
ε_r	$= 1 + \chi_e$
D	$= \varepsilon_0 \varepsilon_r E$
P	$= \varepsilon_0 (\varepsilon_r - 1) E$

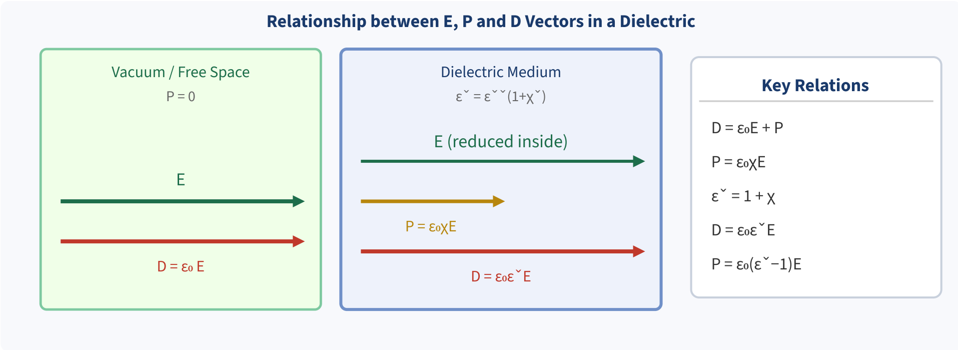

Fig. 3.1 – Relationship between electric field E, polarization P, and displacement vector D inside a dielectric medium.

3.3 Physical Significance

- **E** is the total macroscopic electric field in the material. It includes contributions from free charges and bound charges.
- **P** is the response of the medium – it represents the bound charge contribution and is zero in vacuum.
- **D** depends only on free charges (by Gauss's law: $\nabla \cdot D = \rho_{free}$). It simplifies boundary condition problems.

Note: In vacuum, P = 0, so D = ε_0 E. Inside a linear dielectric, both P and D are proportional to E. The dielectric constant ε_r tells us how much the dielectric reduces the electric field for a given free charge distribution.

SECTION 4

Types of Polarization

Polarization in a dielectric material arises due to four basic mechanisms, each associated with a different physical displacement. The total polarizability is the sum of contributions from all active mechanisms.

TOTAL POLARIZABILITY

$\alpha_{total} = \alpha_e + \alpha_i + \alpha_o + \alpha_s$

α_e = Electronic polarizability

α_i = Ionic polarizability

α_o = Orientational (dipolar) polarizability

α_s = Space charge (interfacial) polarizability

4.1 Electronic Polarization (Quantitative)

Definition: Electronic polarization occurs due to the displacement of the negatively charged electron cloud relative to the positively charged nucleus of an atom under an applied electric field.

Theory and Derivation:

1. Consider a neutral atom with atomic number Z, nucleus charge +Ze, and spherical electron cloud of radius R with uniform charge density $\rho = -3Ze / (4\pi R^3)$.
2. Applied field E displaces the electron cloud by distance x from the nucleus. The nucleus and electron cloud form a dipole.
3. The restoring force on the electron cloud is due to the Coulomb attraction of the nucleus and equals: $F_{restoring} = (Ze^2 / 4\pi\varepsilon_0 R^3)\, x$
4. At equilibrium, the applied force equals the restoring force: $Ze\,E = (Ze^2 / 4\pi\varepsilon_0 R^3)\, x$
5. Solving for x: $x = (4\pi\varepsilon_0 R^3 E) / (Ze)$
6. Induced dipole moment: $p = Ze \times x = 4\pi\varepsilon_0 R^3 E$
7. Since $p = \alpha_e E$, we get: $\alpha_e = 4\pi\varepsilon_0 R^3$

ELECTRONIC POLARIZABILITY

$\alpha_e \quad = 4\pi\varepsilon_0 \; R^3$

$P_e \quad = N \; \alpha_e \; E \; = 4\pi\varepsilon_0 \; N \; R^3 \; E$

α_e is proportional to the volume of the atom (R^3). Larger atoms polarize more easily.

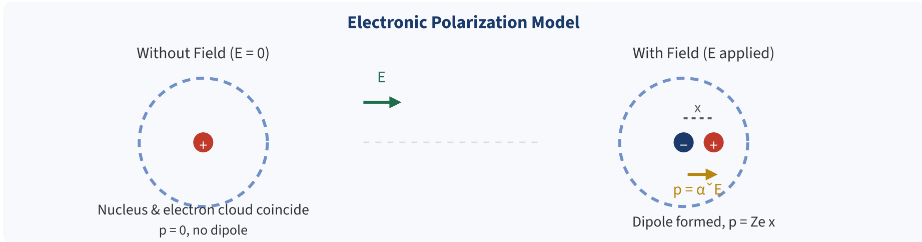

Fig. 4.1 – Electronic polarization: applied field E shifts the electron cloud (blue) relative to nucleus (+), creating dipole moment p = α_e E.

4.2 Ionic Polarization (Quantitative)

Definition: Ionic polarization occurs in ionic crystals (such as NaCl) where the applied electric field causes a relative displacement of positive and negative ion sublattices, creating a net dipole moment.

Derivation:

1. In an ionic crystal, consider a pair of ions with masses M_+ (positive ion) and M_- (negative ion) connected by a bond of effective spring constant β.
2. Applied field E exerts force +eE on positive ion and −eE on negative ion, causing relative displacement x.
3. Restoring force: $F_r = \beta x$ (Hooke's law for the ionic bond)
4. At equilibrium: $eE = \beta x \Rightarrow x = eE / \beta$
5. Induced dipole moment per ion pair: $p_i = e x = e^2 E / \beta$
6. Therefore ionic polarizability: $\alpha_i = e^2 / \beta$

IONIC POLARIZABILITY

$\alpha_i \quad = e^2 / \beta \quad (\beta = \text{bond spring constant})$

$\alpha_i \quad = e^2 / (M_r \, \omega_0^2) \quad (M_r = \text{reduced mass}, \; \omega_0 = \text{resonance frequency})$

α_i is typically 10× smaller than α_e; operative only at low frequencies (< IR range)

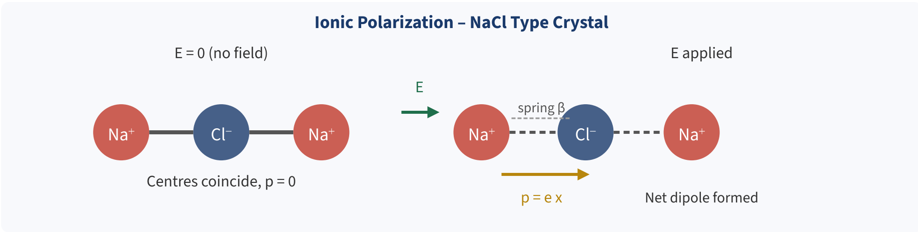

Fig. 4.2 – Ionic polarization in NaCl crystal: applied field E displaces positive (Na^+) and negative (Cl^-) sublattices in opposite directions, producing dipole moment.

4.3 Orientational (Dipolar) Polarization (Qualitative)

Definition: Orientational polarization arises in polar molecules (those with permanent dipole moments, e.g., H_2O, HCl, acetone) when the applied electric field tends to align the randomly oriented permanent dipoles along the field direction.

In the absence of an electric field, the permanent dipoles of polar molecules are randomly oriented due to thermal agitation, so the net dipole moment is zero. When a field E is applied, it exerts a torque $\tau = p \times E$ on each dipole, tending to align them with the field. Thermal motion opposes this alignment.

The degree of alignment is described by the **Langevin function**. At ordinary temperatures and moderate fields ($pE \ll kT$), the orientational polarizability is given by:

ORIENTATIONAL POLARIZABILITY (DEBYE)

$\alpha_o = p_0^2 / (3\ k_B\ T)$ (p_0 = permanent dipole moment, k_B = Boltzmann constant)

α_o decreases with increasing temperature (thermal randomization increases)

Present only in polar materials; absent in non-polar materials

4.4 Comparison of Polarization Types

Property	Electronic	Ionic	Orientational	Space Charge
Mechanism	Electron cloud shift	Ion displacement	Dipole alignment	Charge accumulation at interfaces
Materials	All materials	Ionic crystals	Polar molecules	Inhomogeneous materials
Temperature dependence	Very weak	Weak	Strong (T^{-1})	Moderate
Frequency response	UV / optical	IR ($\approx 10^{13}$ Hz)	Microwave / RF	Low ($< 10^3$ Hz)
Relaxation time	$\approx 10^{-15}$ s	$\approx 10^{-13}$ s	$\approx 10^{-10}$ to 10^{-6} s	$\approx 10^{-2}$ to 1 s
Polarizability magnitude	Smallest	Moderate	Largest (polar)	Very large

Examples	He, H_2, diamond	NaCl, KBr	H_2O, HCl, polar polymers	Ceramics, composites

SECTION 5

Lorentz Internal Field & Clausius–Mossotti Equation

5.1 Lorentz Internal (Local) Field

Definition: The Lorentz internal field (or local field) E_{loc} is the actual electric field acting on a molecule or atom inside a dielectric, which is different from the macroscopic applied field E due to contributions from surrounding polarized molecules.

Macroscopic Maxwell equations give the average field E, but the field actually experienced by an individual molecule inside the dielectric includes contributions from all other polarized molecules. To calculate E_{loc}, Lorentz used a clever construction:

Lorentz Cavity Method:

1. **Applied field E_0**: The external uniform field applied to the dielectric.
2. **Depolarization field $E_1 = -P/\varepsilon_0$**: Field due to the surface bound charges of the entire specimen (opposes E_0).
3. **Lorentz cavity field $E_2 = +P/(3\varepsilon_0)$**: Field due to the surface charges on the spherical Lorentz cavity cut around the molecule (positive contribution).
4. **Molecular field $E_3 = 0$**: For a cubic lattice or isotropic distribution, the field from molecules inside the cavity sums to zero by symmetry.
5. Total local field: $E_{loc} = E_0 + E_1 + E_2 + E_3$
6. Since $E = E_0 + E_1$ (macroscopic field after depolarization), we get:
7. $E_{loc} = E + P/(3\varepsilon_0)$

LORENTZ LOCAL FIELD

$E_{loc} = E + P / (3\varepsilon_0)$

$E_{loc} = (\varepsilon_r + 2) / 3 \times E$ (equivalent form)

$E_{loc} > E$ always: the local field is larger than the macroscopic field

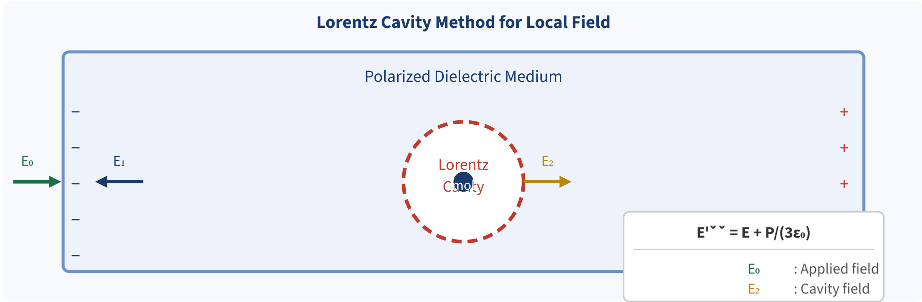

Fig. 5.1 – Lorentz cavity model: the local field E_{loc} at the site of a molecule is the sum of the external field E and the cavity contribution $P/(3\varepsilon_0)$.

5.2 Clausius–Mossotti Equation (Derivation)

The Clausius–Mossotti relation connects the macroscopically measurable dielectric constant ε_r with the microscopic molecular polarizability α. It is valid for non-polar and weakly polar dielectrics.

Derivation:

1. For N dipoles per unit volume, total polarization: $P = N \alpha E_{loc}$
2. Substitute Lorentz field: $P = N \alpha (E + P/(3\varepsilon_0))$
3. Also, $P = \varepsilon_0(\varepsilon_r - 1) E$, so $E = P / (\varepsilon_0(\varepsilon_r - 1))$
4. Substituting into step 2: $P = N \alpha [P/(\varepsilon_0(\varepsilon_r - 1)) + P/(3\varepsilon_0)]$
5. Dividing both sides by P and ε_0:
6. $1 = (N\alpha/\varepsilon_0) [1/(\varepsilon_r - 1) + 1/3]$
7. Rearranging: $1/(N\alpha/\varepsilon_0) = 3/(\varepsilon_r - 1) - 1/3$ ... leads to the final result:

CLAUSIUS–MOSSOTTI EQUATION

$$(\varepsilon_r - 1) / (\varepsilon_r + 2) = N \alpha / (3 \varepsilon_0)$$

Valid for non-polar dielectrics and gases at all frequencies

Optical form (n = refractive index): $(n^2 - 1) / (n^2 + 2) = N \alpha_e / (3 \varepsilon_0)$

Molar refractivity: $R_m = (M/\rho) \times (\varepsilon_r - 1)/(\varepsilon_r + 2) = N_A\alpha / (3\varepsilon_0)$

Key Implication: The Clausius–Mossotti equation enables calculation of microscopic polarizability α from macroscopic measurement of ε_r. It breaks down for strongly polar materials (large α_o) and for ferroelectrics where cooperative effects dominate.

SECTION 6

Frequency Dependence of Polarization & Dielectric Loss

6.1 Frequency Dependence of Polarization

Different polarization mechanisms have different relaxation times and therefore respond differently as the frequency of the applied AC field increases. At low frequencies, all mechanisms contribute. As frequency increases, each mechanism drops out successively as its relaxation time becomes too long for it to follow the field.

Frequency Range	Active Polarizations	ε_r variation
DC to 10^3 Hz	All four: space charge + orientational + ionic + electronic	Highest
10^3 to 10^9 Hz (RF / Microwave)	Ionic + Electronic (orientational drops out)	Decreases
10^9 to 10^{13} Hz (IR)	Electronic only (ionic drops out)	Further decrease
10^{13} to 10^{15} Hz (Visible / UV)	Electronic only	n^2 (optical)
Above UV	None	≈ 1 (vacuum)

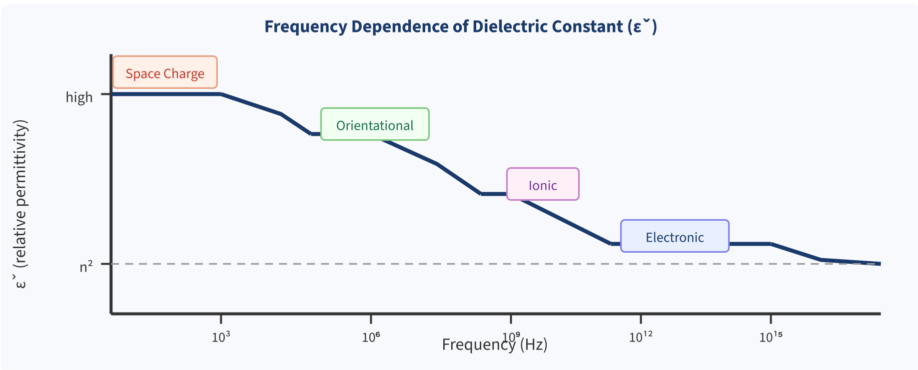

Fig. 6.1 – Variation of dielectric constant ε_r with frequency. Each polarization mechanism becomes inactive above its characteristic resonant/relaxation frequency.

6.2 Complex Dielectric Constant

In a real dielectric subjected to an alternating field, the polarization does not respond instantaneously. This phase lag leads to energy dissipation, represented through a complex dielectric constant:

COMPLEX DIELECTRIC CONSTANT

$\varepsilon^* = \varepsilon' - j\,\varepsilon''$

ε' = Real part (energy storage; related to polarization magnitude)

ε'' = Imaginary part (energy dissipation; dielectric loss)

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tan δ  = ε'' / ε'     (loss tangent or dissipation factor)
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6.3 Dielectric Loss

Dielectric Loss: The energy dissipated as heat in a dielectric material when subjected to an alternating electric field is called dielectric loss. It is caused by the phase lag of polarization behind the applied field.

The power dissipated per unit volume in a dielectric is:

DIELECTRIC LOSS FORMULAS

$P_{loss} = \omega\, \varepsilon_0\, \varepsilon''\, E_0^2 / 2 \quad [\mathrm{W\ m^{-3}}]$

$P_{loss} = \omega\, \varepsilon_0\, \varepsilon'\, (\tan\delta)\, E_0^2 / 2$

P_{loss} increases with frequency ω and loss tangent tan δ

Loss is maximum near relaxation frequencies of each polarization mechanism

Causes of Dielectric Loss:

- Orientation of polar molecules in AC field (dominant loss in polar dielectrics)
- DC conductivity loss: σE^2 — from residual free carriers
- Ionic displacement hysteresis in ionic crystals at IR frequencies
- Resonance absorption at UV / optical frequencies

Applications of Dielectric Loss:

- Microwave cooking: water molecules (permanent dipoles) absorb microwave energy and heat food
- Dielectric heating for welding plastics and drying moist materials
- Low-loss dielectrics (PTFE, silica) used in RF cables and antennas to minimize energy waste
- Tan δ monitoring used to assess insulation condition in high-voltage equipment

SECTION 7

Introduction to Magnetic Materials

7.1 Magnetism – Overview

Magnetism is a fundamental property of matter arising from the motion of electric charges. All materials exhibit some form of magnetic behaviour when placed in an external magnetic field. The origin of magnetism lies in the quantum mechanical properties of electrons: their orbital motion around nuclei and their intrinsic spin.

The response of a material to an applied magnetic field is characterized by its *magnetic susceptibility* χ_m. Based on χ_m, materials are classified as diamagnetic, paramagnetic, ferromagnetic, antiferromagnetic, or ferrimagnetic.

7.2 Atomic Origin of Magnetism

The magnetic moment of an atom arises from three contributions:

- **Orbital magnetic moment (μ_L):** Due to the revolution of electrons around the nucleus. An electron in orbit behaves like a tiny current loop.
- **Spin magnetic moment (μ_S):** Due to the intrinsic quantum-mechanical spin angular momentum of the electron. It has a fixed magnitude of one Bohr magneton.
- **Nuclear magnetic moment:** Due to nuclear spin; approximately 2000 times smaller than electron contributions and usually negligible in bulk magnetism.

BOHR MAGNETON AND ATOMIC MAGNETIC MOMENTS

$\mu_B = e\,h / (4\pi\, m_e) = 9.274 \times 10^{-24}$ A m^2 (Bohr magneton)

$\mu_L = \mu_B \sqrt{l(l+1)}$ (l = orbital quantum number)

$\mu_S = 2\mu_B \sqrt{s(s+1)}$ (s = spin quantum number = 1/2 for one electron)

For one unpaired electron: $\mu_S = \sqrt{3}\,\mu_B$

In filled electron shells and subshells, orbital and spin moments of individual electrons pair up and cancel. Materials with unpaired electrons (transition metals, rare earths) have net magnetic moments and show paramagnetic or ferromagnetic behaviour. Materials with only paired electrons show diamagnetism.

7.3 Magnetic Dipole and its Field

Magnetic Dipole Moment (m): A current-carrying loop of area A carrying current I constitutes a magnetic dipole with moment m = I A. The direction of m is along the axis of the loop, given by the right-hand rule.

MAGNETIC DIPOLE MOMENT

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m      = I A    [SI unit: A m²]
τ      = m × B    (torque on dipole in external field B)
U      = − m · B    (potential energy in field B)
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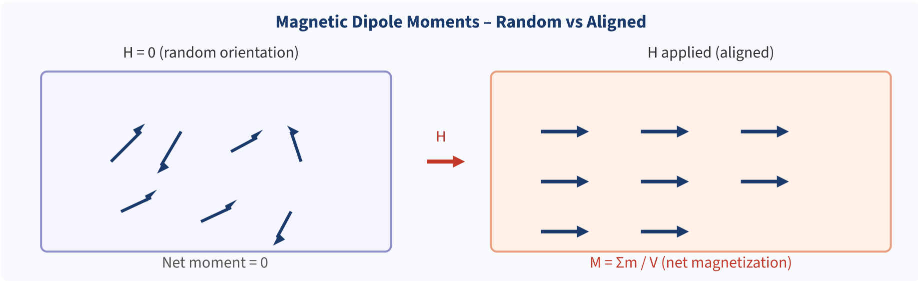


Fig. 7.1 – Magnetic dipole moments in a material: random (H = 0, M = 0) vs aligned along applied field H (M ≠ 0).

SECTION 8

Magnetic Parameters: Magnetization, Susceptibility & Permeability

8.1 Magnetization (M)

Magnetization (M): Magnetization is defined as the net magnetic dipole moment per unit volume of a magnetic material. It represents the degree of alignment of atomic magnetic moments within the material.

MAGNETIZATION

M	$= dm / dV$	(magnetic moment per unit volume)
M	$= \chi_m H$	(for linear magnetic materials)
Unit	$A\ m^{-1}$	

8.2 Magnetic Field Intensity (H)

Magnetic Field Intensity (H): H is the magnetic field produced by free currents (applied sources) alone, independent of the magnetic properties of the medium. It is also called magnetizing force.

MAGNETIC FIELD RELATIONS

H	$= B/\mu_0 - M$	(in a magnetic medium)
B	$= \mu_0(H + M)$	(fundamental relation)
B	$= \mu_0(1 + \chi_m) H = \mu_0 \mu_r H = \mu H$	
μ_r	$= 1 + \chi_m$	
Unit of H	$A\ m^{-1}$	

8.3 Magnetic Flux Density (B)

Magnetic Flux Density (B): B is the total magnetic flux per unit area. It includes contributions from both the applied field H and the magnetization M of the material. B is the physically fundamental field (force on moving charges is $F = q\, v \times B$).

8.4 Magnetic Susceptibility (χ_m)

Magnetic Susceptibility (χ_m): Magnetic susceptibility is a dimensionless quantity that expresses how much a material is magnetized in response to an applied magnetic field H.

MAGNETIC SUSCEPTIBILITY

$$\chi_m = M / H$$

$\chi_m < 0$ (small): Diamagnetic (weakly repels field)

$\chi_m > 0$ (small): Paramagnetic (weakly attracted to field)

$\chi_m \gg 0$ (large, 10^2 to 10^6): Ferromagnetic (strongly attracted)

8.5 Magnetic Permeability (μ)

Magnetic Permeability (μ): Permeability is the measure of the ability of a material to support the formation of a magnetic field within itself. It is the ratio of B to H.

PERMEABILITY

$$\mu = B / H \quad [\text{SI unit: H m}^{-1}]$$

$$\mu_0 = 4\pi \times 10^{-7}\ \text{H m}^{-1} \quad \text{(free space)}$$

$$\mu_r = \mu / \mu_0 = 1 + \chi_m \quad \text{(relative permeability, dimensionless)}$$

$\mu_r < 1$: diamagnetic; $\mu_r \approx 1$: paramagnetic; $\mu_r \gg 1$: ferromagnetic

Quantity	Symbol	Definition	Unit
Magnetic dipole moment	m	m = I A	A m^2
Magnetization	M	M = dm/dV	A m^{-1}
Magnetic field intensity	H	From free currents	A m^{-1}
Magnetic flux density	B	$B = \mu_0(H+M)$	T
Magnetic susceptibility	χ_m	$\chi_m = M/H$	Dimensionless
Relative permeability	μ_r	$\mu_r = 1 + \chi_m$	Dimensionless

SECTION 10

Classification of Magnetic Materials

10.1 Diamagnetic Materials

Definition: Diamagnetic materials are those that are weakly repelled by an external magnetic field. They have no unpaired electrons and develop magnetization opposite to the applied field.

Properties: χ_m is small and negative (-10^{-6} to -10^{-5}); μ_r slightly less than 1; no permanent magnetic moment; effect is temperature-independent; field lines are pushed out of the material.

Origin: Applied field induces change in orbital motion (Lenz's law at atomic level), producing a magnetization opposing H.

Examples: Bismuth (Bi), copper (Cu), silver (Ag), gold (Au), water, nitrogen gas, most organic compounds, superconductors (perfect diamagnets).

Applications: Magnetic levitation in superconductors, precision balance shielding, NMR contrast.

10.2 Paramagnetic Materials

Definition: Paramagnetic materials have atoms with unpaired electrons and a net permanent magnetic moment per atom. They are weakly attracted to external magnetic fields, with magnetization parallel to H.

Properties: χ_m is small and positive (10^{-5} to 10^{-3}); μ_r slightly greater than 1; magnetization disappears when H is removed; strongly temperature-dependent (Curie's law).

CURIE'S LAW (PARAMAGNETISM)

χ_m = C / T (C = Curie constant, T = absolute temperature)

M decreases as T increases — thermal agitation disrupts alignment

Examples: Aluminium (Al), platinum (Pt), manganese (Mn), oxygen (O_2), gadolinium above Curie temperature, most transition metal salts.

10.3 Ferromagnetic Materials

Definition: Ferromagnetic materials exhibit very strong magnetization parallel to the applied field due to quantum exchange interaction between adjacent atoms, which spontaneously aligns neighbouring spins in the same direction.

Properties: χ_m is very large and positive (10^3 to 10^6); $\mu_r \gg 1$; permanent magnetization retained after H is removed (hysteresis); organized into magnetic domains; loses ferromagnetism above Curie temperature T_C.

CURIE–WEISS LAW (FERROMAGNETISM ABOVE T_C)

$$\chi_m = C / (T - T_C) \quad (T > T_C\text{; behaves paramagnetically above Curie point})$$

Below T_C: ferromagnetic; above T_C: paramagnetic

Examples: Iron (Fe), nickel (Ni), cobalt (Co), gadolinium (Gd below 292 K), neodymium alloys (NdFeB).

10.4 Antiferromagnetic Materials

Definition: In antiferromagnetic materials, adjacent atomic magnetic moments are aligned antiparallel and equal in magnitude due to exchange interaction. The net magnetization is zero.

Properties: Small positive susceptibility; below the Néel temperature T_N, alternate layers of spins are antiparallel; above T_N, behaves paramagnetically (Curie–Weiss law with negative θ).

Examples: Manganese oxide (MnO), iron oxide (FeO), chromium (Cr), manganese (Mn) in crystalline form, nickel oxide (NiO).

10.5 Ferrimagnetic Materials (Ferrites)

Definition: Ferrimagnetic materials have two sublattices with antiparallel magnetic moments of different magnitudes, resulting in a net non-zero magnetization. This is a compromise between ferromagnetism and antiferromagnetism.

Properties: High electrical resistivity (reduces eddy current losses); significant net magnetization; magnetic ordering below ferrimagnetic Curie temperature; crystalline structure is typically spinel or garnet.

Examples: Magnetite (Fe_3O_4), ferrites of the type MFe_2O_4 (M = Mn, Zn, Ni, Co), yttrium iron garnet (YIG).

Applications: Transformer cores at high frequencies, microwave devices, recording heads, antenna rods, inductors.

10.6 Summary Classification Table

Property	Diamagnetic	Paramagnetic	Ferromagnetic	Antiferro.	Ferrimagnetic
Susceptibility χ_m	Small, < 0	Small, > 0	Large, >> 0	Small, > 0	Large, > 0
Relative permeability μ_r	< 1	≈ 1 (slightly >1)	≫ 1	≈ 1	> 1
Permanent moment	No	Yes (atomic)	Yes (domain)	No (net = 0)	Yes (net ≠ 0)
Spin alignment	None	Random	Parallel	Antiparallel (equal)	Antiparallel (unequal)
Temperature dependence	None	$\chi \propto T^{-1}$	Curie–Weiss	Néel T_N	Curie T_C

Examples	Bi, Cu, Au	Al, Pt, Mn	Fe, Ni, Co	MnO, NiO	Fe_3O_4, ferrites

SECTION 11

Domain Theory of Ferromagnetism

11.1 Concept of Magnetic Domains

Magnetic Domain: A magnetic domain is a microscopic region within a ferromagnetic material in which all atomic magnetic moments are aligned parallel due to quantum mechanical exchange interaction. Within each domain the material is spontaneously and completely magnetized, but adjacent domains may have different orientations.

The existence of domains explains the apparent paradox: pure iron is ferromagnetic (neighbouring spins parallel), yet an unmagnetized piece of iron has no macroscopic magnetic moment. This is because randomly oriented domains cancel each other out, giving $M = 0$ overall even though each domain is saturated.

11.2 Origin and Energetics of Domains

Domain formation minimizes the total free energy of the ferromagnet, which includes:

- **Exchange energy:** Drives adjacent spins to be parallel (quantum mechanical; minimized within each domain).
- **Magnetostatic (stray field) energy:** Driving force to break into multiple domains; field lines returning through air increase this energy for a single-domain crystal.
- **Magnetocrystalline anisotropy energy:** Tendency of magnetization to align along preferred crystallographic axes (easy axes).
- **Magnetostrictive (elastic) energy:** Lattice strains caused by magnetization; can be reduced by closure domains.
- **Domain wall energy:** Energy stored in the transition region (wall) between adjacent domains. Opposes further subdivision.

The equilibrium domain size results from a balance between the decrease in magnetostatic energy (favoring more/smaller domains) and the increase in domain wall energy (favoring fewer/larger domains).

11.3 Domain Walls (Bloch Walls)

Domain Wall (Bloch Wall): The transition region between two adjacent magnetic domains in which the direction of magnetization gradually changes from that of one domain to the other. The width of a typical Bloch wall is 100–300 lattice spacings (≈ 30–100 nm).

In a Bloch wall, the magnetization rotates in a plane perpendicular to the wall plane. The wall width δ is determined by the balance between exchange energy (favoring wide walls, gradual rotation) and anisotropy energy (favoring narrow walls, abrupt rotation):

DOMAIN WALL PROPERTIES

$\delta = \pi \sqrt{(A / K)}$ (A = exchange stiffness, K = anisotropy constant)

$\sigma_w = 4 \sqrt{(A K)}$ (wall energy per unit area)

Typical: $\delta \approx 50\text{–}500$ nm; $\sigma_w \approx 0.001\text{–}0.1\ \mathrm{J\ m^{-2}}$

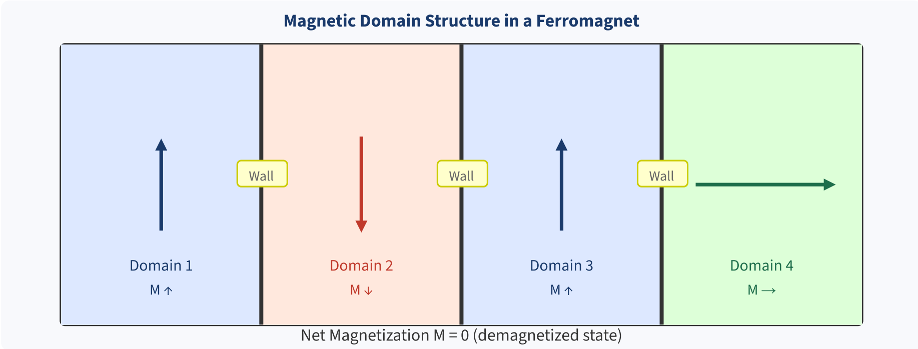


Fig. 11.1 – Domain structure of a demagnetized ferromagnetic crystal. Adjacent domains have different magnetization orientations; net moment is zero. Domain walls (Bloch walls) separate them.

11.4 Magnetization Process (Domain Model)

When an external field H is applied to a demagnetized ferromagnet, the following stages occur:

- **Stage 1 – Reversible wall motion:** Domains with magnetization close to H grow at the expense of unfavorably oriented domains by small, reversible wall displacements. (Low H region of B–H curve)
- **Stage 2 – Irreversible wall motion:** At higher H, domain walls jump over lattice defects and impurities (Barkhausen jumps). This is irreversible — energy is lost as heat. (Steep region of B–H curve)
- **Stage 3 – Domain rotation:** At very high H, all walls have moved and single-domain regions rotate their magnetization to align exactly with H. This requires overcoming anisotropy energy. (Near-saturation region)
- **Saturation:** All domains aligned with H; $M = M_s$ (saturation magnetization).

SECTION 12

Hysteresis and Magnetic Materials

12.1 The B–H Hysteresis Loop

Hysteresis: The lagging of magnetization (or B) behind the magnetizing force H when H is cycled is called hysteresis. The closed loop traced on a B–H graph through a complete magnetization cycle is the hysteresis loop. The area of the loop represents energy dissipated as heat per unit volume per cycle.

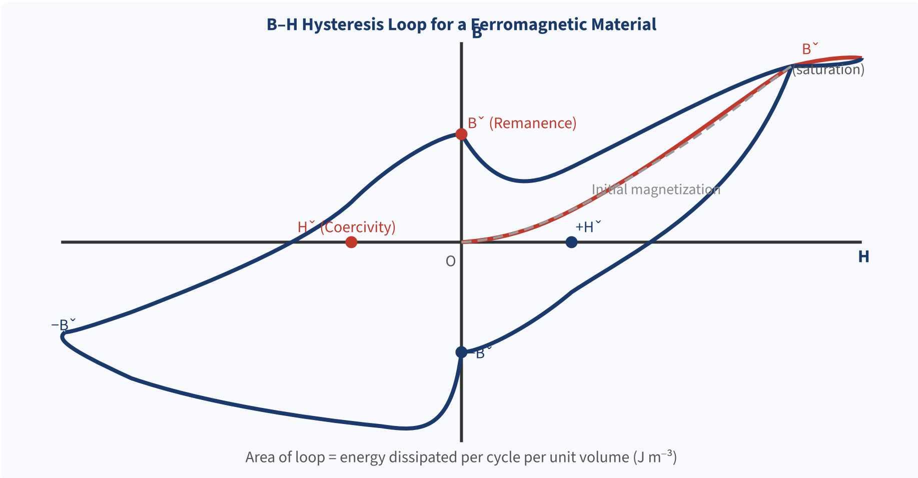


Fig. 12.1 – B–H hysteresis loop. B_r = remanence; H_c = coercive field; B_s = saturation flux density. The loop area gives hysteresis loss per cycle per unit volume.

12.2 Important Parameters of the Hysteresis Loop

HYSTERESIS LOOP PARAMETERS

B_s = Saturation flux density: maximum B at full domain alignment

B_r = Remanence (residual induction): B remaining when H returns to zero after saturation

H_c = Coercive field (coercivity): H needed to reduce B to zero after saturation

W_h = Hysteresis loss per cycle = area of B–H loop [J m^{-3}]

P_h = Steinmetz: $P_h = \eta\, B_s^n\, f$ (η = Steinmetz coefficient, $n \approx 1.6$, f = frequency)

12.3 Soft and Hard Magnetic Materials

SOFT MAGNETIC MATERIALS

Definition: Soft magnetic materials are easily magnetized and demagnetized. They have narrow hysteresis loops (small H_c, small B_r), indicating low hysteresis loss per cycle.

Properties: Low coercivity H_c; high permeability μ_r; low hysteresis loss; small loop area; quickly respond to changing H; no permanent magnetism retained.

Examples: Soft iron (Fe), silicon steel (Fe–Si), Permalloy (78% Ni–22% Fe), Mumetal, ferrites (Mn–Zn, Ni–Zn).

Applications: Transformer cores, electromagnet cores, motor and generator cores, inductors, magnetic shields, recording read-heads.

HARD MAGNETIC MATERIALS

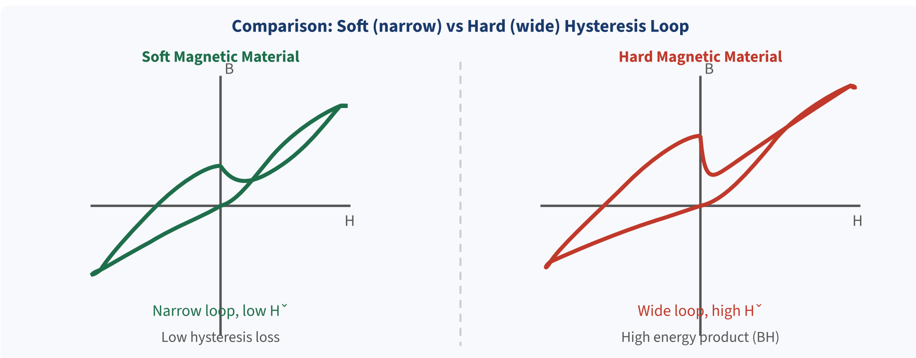
Definition: Hard magnetic materials are difficult to magnetize and demagnetize. They have wide hysteresis loops (large H_c, large B_r), storing large amounts of magnetic energy.

Properties: High coercivity H_c; high remanence B_r; large BH_{max} product (energy product); retain strong permanent magnetism; wide loop area.

Examples: Alnico alloys (Al–Ni–Co), hard ferrites ($BaFe_{12}O_{19}$), neodymium magnets ($Nd_2Fe_{14}B$), SmCo alloys.

Applications: Permanent magnets, loudspeakers, electric motors (PM type), magnetic storage (HDDs), MRI machines, sensors, generators.

Property	Soft Magnetic	Hard Magnetic
Coercivity H_c	Low (< 1000 A m^{-1})	High (> 10^4 to 10^6 A m^{-1})
Remanence B_r	Low to moderate	High
Permeability μ_r	Very high (10^3–10^5)	Moderate to low
Hysteresis loop	Narrow (small area)	Wide (large area)
Energy loss per cycle	Low	High
Magnetization ease	Easy	Difficult
Demagnetization ease	Easy	Very difficult
BH_{max} product	Low	High
Main use	AC electromagnetic cores	Permanent magnets, storage
Examples	Silicon steel, Permalloy, MnZn ferrite	Alnico, $Nd_2Fe_{14}B$, $BaFe_{12}O_{19}$



max

Fig. 12.2 – Left: narrow hysteresis loop of soft magnetic material (low coercivity, low loss). Right: wide loop of hard magnetic material (high coercivity, high remanence, used for permanent magnets).

SECTION 13

Questions & Answers

Part A – 2 Mark Questions

Q1. Define dielectric polarization and polarizability.
2 Marks

Dielectric Polarization (P): It is the net electric dipole moment per unit volume induced in a dielectric material by an applied electric field. Mathematically, $P = N p$, where N is the number density of molecules and p is the average induced dipole moment per molecule. Unit: $C\ m^{-2}$.

Polarizability (α): It is the proportionality constant between the induced dipole moment p of a molecule and the local electric field E_{loc} acting on it. $p = \alpha E_{loc}$. Unit: $F\ m^2$. A larger α means the molecule is more easily polarized.

Q2. What is the dielectric constant? Give its physical significance.
2 Marks

Dielectric Constant (ε_r): It is the ratio of the permittivity ε of the dielectric material to the permittivity ε_0 of free space: $\varepsilon_r = \varepsilon / \varepsilon_0$.

Physical Significance: (i) $\varepsilon_r = C / C_0$: insertion of dielectric increases capacitance ε_r times. (ii) ε_r represents the factor by which the electric field is reduced inside a dielectric compared to vacuum for the same free charge. (iii) $\varepsilon_r \geq 1$ always. Vacuum: $\varepsilon_r = 1$. Water: $\varepsilon_r \approx 80$. $BaTiO_3$: ε_r up to 1500.

Q3. Define magnetic susceptibility and relative permeability.
2 Marks

Magnetic Susceptibility (χ_m): It is the ratio of magnetization M to the applied magnetic field intensity H: $\chi_m = M/H$. It is dimensionless. Positive χ_m indicates paramagnetic/ferromagnetic; negative χ_m indicates diamagnetic behaviour.

Relative Permeability (μ_r): It is the ratio of the absolute permeability μ of the material to the permeability of free space μ_0: $\mu_r = \mu/\mu_0$. It is dimensionless and related to susceptibility by $\mu_r = 1 + \chi_m$.

Q4. What is the Lorentz local field and how does it differ from the macroscopic field?
2 Marks

Lorentz Local Field (E_{loc}): It is the actual electric field experienced by a single molecule inside a dielectric, calculated by Lorentz using the cavity method: $E_{loc} = E + P/(3\varepsilon_0)$, where E is the macroscopic average field and P is the polarization.

Difference: The macroscopic field E is the spatial average of the microscopic field. The local field E_{loc} is larger than E because it includes the field from the nearby polarized molecules (cavity correction term $P/(3\varepsilon_0)$ is positive). For dense dielectrics, using E instead of E_{loc} would significantly underestimate the true polarizing force on each molecule.

Q5. Define dielectric loss and loss tangent. **2 Marks**

Dielectric Loss: When a dielectric is subjected to an alternating electric field, the polarization cannot follow the field instantaneously. The phase lag between P and E results in energy being absorbed from the field and dissipated as heat. This energy dissipation is called dielectric loss.

Loss Tangent (tan δ): It is the ratio of the imaginary part to the real part of the complex dielectric constant: $\tan\delta = \varepsilon'' / \varepsilon'$. It measures the fraction of energy lost per cycle. For good insulators $\tan\delta < 10^{-4}$; for high-loss materials $\tan\delta \approx 0.01–0.1$. The dissipation power is $P = \omega\, \varepsilon_0\, \varepsilon' (\tan\delta)\, E_0^2 / 2$.

Q6. Distinguish between soft and hard magnetic materials. **2 Marks**

Soft Magnetic Materials: Easily magnetized and demagnetized; narrow hysteresis loop; low coercivity ($H_c < 1000$ A m^{-1}); high permeability; low hysteresis loss. Used in transformer cores, inductors, electromagnets. Examples: silicon steel, Permalloy.

Hard Magnetic Materials: Difficult to magnetize and demagnetize; wide hysteresis loop; high coercivity ($H_c > 10^4$ A m^{-1}); high remanence; retain permanent magnetism. Used in permanent magnets, loudspeakers, magnetic storage. Examples: Alnico, $Nd_2Fe_{14}B$, $BaFe_{12}O_{19}$.

Part B – 5 Mark Questions

Q7. Explain the types of polarization in dielectrics in detail. **5 Marks**

Polarization in dielectrics occurs due to four mechanisms:

1. Electronic Polarization: When an electric field E is applied, the electron cloud of each atom is displaced relative to the nucleus, creating an induced dipole. This occurs in all materials. Electronic polarizability $\alpha_e = 4\pi\varepsilon_0 R^3$ (R = atomic radius). This is the fastest mechanism, responding up to optical/UV frequencies ($\approx 10^{15}$ Hz). It is temperature-independent.

2. Ionic Polarization: Present only in ionic crystals (NaCl, KBr etc.). The applied field displaces the positive and negative ion sublattices in opposite directions. Ionic polarizability $\alpha_i = e^2/(\beta)$, where β is the effective spring constant of the ionic bond. Active up to IR frequencies ($\approx 10^{13}$ Hz). Slightly temperature-dependent.

3. Orientational (Dipolar) Polarization: Occurs only in polar molecules (H_2O, HCl) which have permanent dipole moments. Without E, dipoles are randomly oriented; the field exerts a torque aligning them. Described by Langevin function; at moderate fields $\alpha_o = p_0^2/(3k_BT)$. Active up to microwave frequencies ($\approx 10^9$ Hz). Strongly temperature-dependent ($\alpha_o \propto T^{-1}$).

4. Space Charge (Interfacial) Polarization: At interfaces of inhomogeneous materials or near grain boundaries, free charges accumulate under an applied field, building up macroscopic dipole moments. Active only at very low frequencies ($< 10^3$ Hz). Important in ceramics and composites.

Total polarizability: $\alpha = \alpha_e + \alpha_i + \alpha_o + \alpha_s$. As frequency increases, successive mechanisms drop out, causing step-wise decreases in ε_r.

Q8. Derive the Clausius–Mossotti equation. **5 Marks**

Given: N = number of molecules per unit volume, α = molecular polarizability, ε_r = dielectric constant.

Step 1 – Lorentz local field: For a cubic / isotropic dielectric, the local field is $E_{loc} = E + P/(3\varepsilon_0)$.

Step 2 – Polarization expression: Total polarization $P = N \alpha E_{loc} = N \alpha [E + P/(3\varepsilon_0)]$ (i)

Step 3 – Express E in terms of P: From $P = \varepsilon_0(\varepsilon_r - 1)E$, we get $E = P/[\varepsilon_0(\varepsilon_r-1)]$ (ii)

Step 4 – Substitute (ii) into (i): $P = N\alpha [P/(\varepsilon_0(\varepsilon_r-1)) + P/(3\varepsilon_0)]$

Step 5 – Divide both sides by P and ε_0: $1/(N\alpha) = (1/\varepsilon_0) [1/(\varepsilon_r-1) + 1/3]$

Step 6 – Rearranging: $\varepsilon_0/(N\alpha) = 3/(\varepsilon_r-1) \times 1 + 1/3$ ... which simplifies to:
$N\alpha/(3\varepsilon_0) = [(\varepsilon_r-1)/(\varepsilon_r+2)]$ — the Clausius–Mossotti Equation.

Result: $(\varepsilon_r - 1) / (\varepsilon_r + 2) = N\alpha / (3\varepsilon_0)$

Physical Significance: This relates the macroscopic quantity ε_r (measurable in capacitor experiments) to the microscopic quantity α (molecular-scale). It shows that ε_r increases when N (density) or α increases. It breaks down for polar dielectrics where dipole–dipole interactions are important.

Q9. Explain dielectric loss and the complex dielectric constant. Discuss its practical significance. **5 Marks**

Complex Dielectric Constant: When a sinusoidal electric field $E = E_0 e^{j\omega t}$ is applied to a real dielectric, the polarization lags behind the field by a phase angle δ (loss angle). This is represented by a complex dielectric constant: $\varepsilon^* = \varepsilon' - j\varepsilon''$, where ε' is the real (storage) part and ε'' is the imaginary (loss) part.

Loss Tangent: $\tan \delta = \varepsilon'' / \varepsilon'$ is called the dissipation factor or loss tangent. It quantifies the fraction of energy lost per cycle relative to the energy stored.

Dielectric Loss: Power dissipated per unit volume = $P = \omega \varepsilon_0 \varepsilon' \tan \delta E_0^2 / 2$. This increases linearly with frequency ω and with $\tan \delta$.

Mechanisms: (i) Orientation loss: polar molecules lagging behind the field (peak at microwave freq.); (ii) DC conductivity loss from residual charge carriers; (iii) Ionic resonance absorption near IR.

Practical Significance: (i) Microwave ovens exploit dielectric loss of water molecules to heat food. (ii) Low-loss materials (PTFE, silica) are used for RF/microwave transmission lines. (iii) $\tan \delta$ is used as a quality index for cable insulation; rising $\tan \delta$ signals insulation ageing. (iv) Dielectric heating (RF welding) uses high-loss polymers to join plastic sheets without adhesives.

Temperature effect: Loss peak shifts to higher frequency as temperature rises (faster relaxation). Multiple overlapping peaks from different mechanisms are seen in complex spectra.

Q10. Classify magnetic materials with properties and examples. 5 Marks

Magnetic materials are classified into five types based on the nature and magnitude of their magnetic susceptibility χ_m:

1. Diamagnetic: χ_m is small and negative (-10^{-6}). No unpaired electrons; induced magnetization opposes H. $\mu_r < 1$. Temperature-independent. Examples: Bi, Cu, Au, water. Used in magnetic shielding.

2. Paramagnetic: χ_m small and positive (10^{-5} to 10^{-3}). Unpaired electrons; random thermal orientation reduced by H. Curie law: $\chi_m = C/T$. μ_r slightly > 1. Examples: Al, Pt, O_2, Mn salts.

3. Ferromagnetic: χ_m very large (10^3–10^6). Spontaneous parallel spin alignment by exchange interaction; organized into domains. Shows hysteresis. Curie–Weiss law above T_C. $\mu_r \gg 1$. Examples: Fe, Ni, Co, Gd. Used in motors, transformers.

4. Antiferromagnetic: Adjacent spins antiparallel and equal; net M = 0. Small positive χ_m. Below Néel temperature T_N. Examples: MnO, FeO, Cr. Used in exchange-bias systems (hard disks).

5. Ferrimagnetic: Two sublattices with antiparallel but unequal moments; net $M \neq 0$. High resistivity reduces eddy current losses. Below Curie temperature. Examples: Fe_3O_4, ferrites MFe_2O_4. Used in transformers, microwave devices.

Key relation connecting all: $B = \mu_0(H + M)$; $M = \chi_m H$; $\mu_r = 1 + \chi_m$.

Q11. Explain the hysteresis loop (B–H curve) and its significance. Differentiate between soft and hard magnetic materials with examples. 5 Marks

Hysteresis Loop: When H is cycled between $+H_{max}$ and $-H_{max}$, the B–H graph traces a closed loop due to irreversible domain wall motion. Key features:

(i) **Saturation (B_s):** All domains aligned with H; maximum B.
(ii) **Remanence (B_r):** Residual B when H is reduced to zero after saturation; represents retained magnetism.
(iii) **Coercivity (H_c):** Reverse field needed to reduce B to zero; measure of resistance to demagnetization.
(iv) **Loop area = W_h = hysteresis loss** per unit volume per cycle ($J\ m^{-3}$). Total power loss $P_h = W_h \times f$, where f = frequency.

Significance: Hysteresis loss heats transformer cores; designing low-loss cores (silicon steel) reduces energy waste. Hard materials with large loop area store energy as permanent magnets.

Soft vs Hard Materials:
Soft magnetic (Fe, silicon steel, Permalloy): narrow loop, low H_c (< 1000 A m^{-1}), high μ_r, low loss. Used for transformer/motor cores where repeated cycling occurs.
Hard magnetic (Alnico, $Nd_2Fe_{14}B$): wide loop, high H_c (> 10^5 A m^{-1}), high B_r, high $(BH)_{max}$. Used for permanent magnets in motors, loudspeakers, MRI, hard disk drives.

Q12. Explain the domain theory of ferromagnetism and the magnetization process.

5 Marks

Domain Theory (Weiss 1907): A ferromagnetic material below the Curie temperature T_C is divided into small regions called magnetic domains, each spontaneously and completely magnetized. The magnetization of different domains is in different directions, so the net magnetization of the entire sample can be zero in the unmagnetized state.

Origin of Domains: Domains form to minimize total free energy. A single-domain crystal has large magnetostatic (stray field) energy. By subdividing into antiparallel domains, this energy is reduced. The formation is limited by domain wall energy (which increases with more walls). Equilibrium domain size results from balancing these energies.

Domain Walls (Bloch Walls): The transition region ($\delta \approx 50–300$ nm) between adjacent domains where the magnetization gradually rotates from one domain orientation to another. Wall width is determined by competition between exchange energy (favoring gradual rotation) and anisotropy energy (favoring abrupt change).

Magnetization Process (three stages):

1. *Reversible wall motion (low H):* Favorably oriented domains grow via small, reversible Bloch wall movements. Linear B–H region.
2. *Irreversible wall motion (medium H):* Walls jump over lattice imperfections (Barkhausen effect). Steep, irreversible B–H region with hysteresis.
3. *Domain rotation (high H):* Remaining unfavorably oriented domains rotate toward H. Leads to saturation M_s.

Upon removal of H, domain walls do not fully return due to pinning at defects, resulting in remanence B_r. This is the origin of permanent magnetism and hysteresis.

SECTION 14

Quick Revision Sheet

ELECTRIC FIELD RELATIONS

$P = \varepsilon_0 \chi_e E$

$D = \varepsilon_0 E + P$

$D = \varepsilon_0 \varepsilon_r E$

$\varepsilon_r = 1 + \chi_e$

$P = \varepsilon_0(\varepsilon_r - 1) E$

MAGNETIC FIELD RELATIONS

$M = \chi_m H$

$B = \mu_0(H + M)$

$B = \mu_0 \mu_r H$

$\mu_r = 1 + \chi_m$

$\mu_0 = 4\pi\times10^{-7}$ H m^{-1}

POLARIZABILITY TYPES

Type	α	Freq. limit
Electronic	$4\pi\varepsilon_0 R^3$	10^{15} Hz (UV)
Ionic	e^2/β	10^{13} Hz (IR)
Orientational	$p_0^2/(3k_BT)$	10^9 Hz (MW)
Space charge	Large	10^3 Hz

MAGNETIC CLASSIFICATION

Type	χ_m	μ_r
Diamagnetic	-10^{-6}	< 1
Paramagnetic	$+10^{-5}$	≈ 1+
Ferromagnetic	$+10^3$–10^6	≫ 1
Antiferro.	Small +ve	≈ 1+
Ferrimagnetic	$+10 – 10^4$	> 1

KEY EQUATIONS

Lorentz field: $E_{loc} = E + P/(3\varepsilon_0)$

Clausius-Mossotti: $(\varepsilon_r-1)/(\varepsilon_r+2) = N\alpha/(3\varepsilon_0)$

Curie law: $\chi_m = C/T$

Curie-Weiss: $\chi_m = C/(T-T_C)$

Bohr magneton: $\mu_B = 9.274\times10^{-24}$ A m^2

HYSTERESIS LOOP PARAMETERS

- **B_s**: Saturation flux density
- **B_r**: Remanence (residual flux)
- **H_c**: Coercive field
- **Loop area**: Hysteresis loss / cycle
- Soft: narrow loop, low H_c, high μ_r
- Hard: wide loop, high H_c, high B_r

DIELECTRIC LOSS

DOMAIN THEORY SUMMARY

- Domains: regions of uniform spin alignment

$$\varepsilon^* = \varepsilon' - j\varepsilon''$$

$$\tan\delta = \varepsilon'' / \varepsilon'$$

$$P_{loss} = \omega\varepsilon_0\varepsilon'(\tan\delta)E_0^2/2$$

Loss peak near each relaxation freq.

- Bloch wall: 50–500 nm transition region
- Magnetization stages: (1) reversible wall motion, (2) irreversible jumps, (3) domain rotation
- Barkhausen effect: sudden domain wall jumps
- Above T_C: domains disappear, paramagnetic

Important Points to Remember

DIELECTRICS: P always in same direction as E (in isotropic media). D depends only on free charges. $\varepsilon_r = 1 + \chi_e$ always. Clausius–Mossotti is valid only for non-polar dielectrics. Electronic polarization exists in all materials and operates at highest frequency. Orientation polarization is only in polar materials and is strongly temperature-dependent.

MAGNETIC MATERIALS: $B = \mu_0(H + M)$ is universally valid. $\mu_r = 1 + \chi_m$. Ferromagnets lose their properties above the Curie temperature (become paramagnetic). The BH product (energy product) is the figure of merit for permanent magnets. Ferrites have high resistivity, reducing eddy currents — ideal for high-frequency applications.

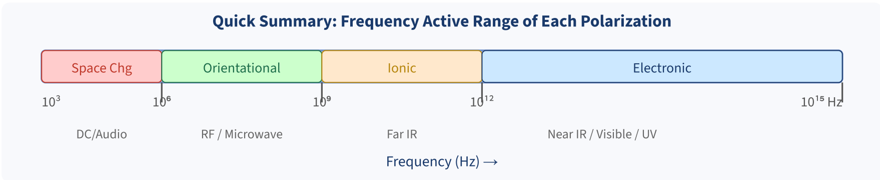


Fig. 14.1 – Active frequency ranges of the four polarization mechanisms.

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Prescribed Textbooks

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2. D.K. Bhattacharya and Poonam Tandon, *Engineering Physics*, Oxford University Press, New Delhi, 2015. *[Chapters 7–9: Dielectric Properties, Magnetic Properties]*

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1. B.K. Pandey and S. Chaturvedi, *Engineering Physics*, Cengage Learning, 2021. *[Comprehensive treatment of polarization mechanisms and ferromagnetism]*
2. Shatendra Sharma and Jyotsna Sharma, *Engineering Physics*, Pearson Education India, 2018. *[Clear derivations of Clausius–Mossotti and hysteresis loop analysis]*
3. Sanjay D. Jain, G.G. Sahasrabudhe, and Girish Joshi, *Engineering Physics*, Universities Press (Orient Blackswan), Hyderabad, 2010.
4. M.R. Srinivasan, *Physics for Engineers*, New Age International Publishers, 2009.

Additional Online Resources

- NPTEL Video Lectures: Engineering Physics – IIT Roorkee / IIT Bombay (nptel.ac.in)
- Hyperphysics: Dielectric Materials – Georgia State University (hyperphysics.phy-astr.gsu.edu)
- MIT OpenCourseWare: 3.024 Electronic, Optical and Magnetic Properties of Materials

Formula Master List

COMPLETE FORMULA REFERENCE – UNIT III

$p = q\,d$ Electric dipole moment [C m]

$P = N\,p = \varepsilon_0\chi_e E$ Polarization [C m^{-2}]

$\alpha_e = 4\pi\varepsilon_0 R^3$ Electronic polarizability

$\alpha_i = e^2/\beta$ Ionic polarizability

$\alpha_o = p_0^2/(3k_BT)$ Orientational polarizability

$D = \varepsilon_0 E + P = \varepsilon_0\varepsilon_r E$ Displacement vector [C m^{-2}]

$\varepsilon_r = 1 + \chi_e$ Relative permittivity

$E_{loc} = E + P/(3\varepsilon_0)$ Lorentz local field

$(\varepsilon_r-1)/(\varepsilon_r+2) = N\alpha/(3\varepsilon_0)$ Clausius-Mossotti equation

$\varepsilon^* = \varepsilon' - j\varepsilon''$ Complex dielectric constant

$\tan\delta = \varepsilon''/\varepsilon'$ Loss tangent

$m = I A$ Magnetic dipole moment [A m^2]

$\mu_B = 9.274\times10^{-24}$ A m^2 Bohr magneton

$M = \chi_m H$ Magnetization [A m^{-1}]

$B = \mu_0(H + M) = \mu_0\mu_r H$ Magnetic flux density [T]

$\mu_r = 1 + \chi_m$ Relative permeability

$\chi_m = C/T$ Curie law (paramagnets)

$\chi_m = C/(T-T_C)$ Curie-Weiss law (above T_C)

W_h = area of B-H loop Hysteresis energy loss [J m^{-3} $cycle^{-1}$]

$\mu_0 = 4\pi\times10^{-7}$ H m^{-1} Permeability of free space

$\varepsilon_0 = 8.854\times10^{-12}$ F m^{-1} Permittivity of free space

End of Unit III – Dielectric & Magnetic Materials

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